Agile Project Management Case Study and Exercises

Based on Handbook v 1.1
(Version 1.3)

Agile Project Management Exercises

Unless otherwise stated, throughout these exercises assume:

- Your group can add any information about the case study not provided (or ask your trainer)
- The exercises may be stopped part way through if your trainer has seen that the learning points have been sufficiently achieved.
- You are going to be the project manager for this project if / when it takes place.
  You work for a small consultancy in the town near to Miller's Farm.

Elevator Pitch Exercise

(An Elevator Pitch is a short overview of the main points, delivered as if you have a key person for a few minutes in an elevator.)

Working in small groups:

- Create a short project objective statement: describing why the project detailed below is taking place (to explain to the CEO of your company in a couple of minutes)
- From what you have learnt so far,
  create a short presentation from the views in your group on:
  - the major benefits and risks of running this project in an agile way
    (Note: here we are not looking for the project risks/benefits which would be true for any approach. Consider here the risks of using Agile.)
  - With what you know about Agile up to now, would your group run this project using an Agile Project Management approach?

  ***Exercise objectives**:*

  1. *Using Agile often involves a PM "selling" the approach to be taken. This works best when supported by a balanced assessment of benefits and risks, rather than just enthusiasm (although enthusiasm is good!).*
  2. *It is a good practice, at the outset of a project, to summarise in a short, succinct way, what the project is about, to help to promote a unified vision of the objectives and direction of the project. This is particularly useful in Agile to keep the project on track and in line with the strategic goal.*

Case Study – Millers Marmalade

History and Background

The Miller family are farmers, and three years ago they successfully expanded into a new area: making and selling of a wide range of marmalades (citrus fruit jams – typically orange jam). They have been so successful that Food World (FW), an international food company has recently taken a 20% share in Millers Marmalade. Although this has helped provide capital and advice on accessing international markets, the parent company has left Millers reasonably independent and in charge of running their own business.

Millers sell most of their marmalade to national supermarkets ('trade sales'). This is the core of their business and is handled by account managers who have built up good relationships with several large customers. They currently use informal systems such spreadsheets and simple database applications but this is now inefficient and results in mistakes being made as the trade volumes are growing rapidly. They now want to expand the sales of marmalade more widely to the general public.

The general public can already buy marmalade from a very basic website that Millers run and maintain. However, supporting and enhancing the website is now causing problems for Millers.

Millers have converted several farm buildings to provide facilities to create large volumes of marmalade. Some of their processes are fully automated such as the making of the marmalade, temperature controlled storage and the cleaning and filling of the marmalade jars.

Millers employ 27 staff who run the machinery, customer service and distribution. They have recently outsourced most of their delivery functions to national transport contractors and to local delivery companies.

Last year Millers made a profit of 50,000 which pleased the directors as they had invested heavily in the marmalade side of the business since it launched three years ago.

Award Winners

Recently Millers Marmalade won the special "Orange d'Or" award at an international marmalade festival in France and the publicity this created has led to a significant rise in orders and enquiries. Everyone at Millers has been very excited about the award and the directors of Millers have decided that they can harness the staff enthusiasm to set another target with a deadline in 6 months time: to get Millers listed on the "100 Best Food Websites" which is a very popular national television (TV) programme. Achieving this could double the demand for Millers Marmalade within months, (specifically Millers would aim to double sales within 2 months following the TV programme)

Work to be done

If Millers go forward with the idea of being listed in the “100 Best Food Websites” the current work they have identified is:

- A significant upgrade to the existing website (or even a new website) in order to offer more advanced features to engage with their customers
- A complete rebranding of the Millers Marmalade image
- Creation of a larger office by moving from the current office: from their small farm building to one of the much larger ones.
- Recruitment of 6 more staff to help with the increased workload
- A marketing campaign to take advantage of the success of the recent award

From initial research done it seems that Millers could rebrand for 20,000. The office move and refit would cost 35,000. A recruitment campaign including a local recruitment event would cost 18,000 (assuming a percentage fee to be paid for 6 staff, based on an average salary of 20,000 per annum). A local company would be engaged for the recruitment.

Costs for the website are 24,000 to build, plus a support contract of 400 per month. Initial estimates suggest that the profit from increased sales could pay for the cost of the website within a few months.

It is critical that the efficiency of the existing website, processes and systems are not affected while any changes are made.



Roles Exercise –
Identifying people for the Business Roles

Working in small groups:

Using the role briefs provided:

- List the best person to fulfil each of the 4 **business** roles
- Make a note of your reasons and any risks with your selections.

- You may be asked by your trainer to consider other DSDM roles too

Be prepared to feedback your group's views to the others, including:

- whether or not you would combine any roles and why
- whether you would assign more than one person to a particular role and why.

***Exercise objectives**:*

To help Agile Project Managers understand the benefits of having the most suitable person in the business roles, be able to recognise the risks of having less suitable people and find the best fit for the roles from the people available.. Getting the right people in the business roles is one of the most important factors contributing to the success of an Agile project.

Roles Exercise – Millers Marmalade

PEOPLE PROFILES FOR THE PROJECT TEAM

Mrs. Miller

Strong accountancy skills. She authorised the 'Top 100' project. Responsible for the running of the farm. Very decisive. Doesn't like the impersonal nature of the internet.

Mr. Miller

Likes to play around with ideas. Starts many things but rarely finishes anything. Decides where the family business should be going at a strategic level. Very keen on technology and technology gadgets.

Jake Miller

'Hands-on' day to day running of the marmalade business. Leads a small team. Is very operationally focussed on getting the right thing to the right customer. Takes most of the telephone calls that do not go directly to the account managers. Very reactive – responds to problems when they have happened.

Kerry Miller

Handles Customer Service and Marketing. Tries to keep everyone happy. Responsible for dealing with any queries including those about the machinery in the marmalade process. Responsible for the 'image' and reputation of Millers Marmalade. Cares passionately about environmental issues.

Victor Popov

The I.T.Manager who started as a website designer at Millers three years ago. Knows all of the existing systems and applications in detail, works long hours and leaves his phone on all the time in case of problems. Likes to straighten the pictures that hang in the reception area so that they look neat and tidy.

Sanjay Patel

Responsible for looking after the team of account managers. His team are involved in selling, and organising the companies who deliver the marmalade (order fulfilment). Very good inter-personal skills. Really loves marmalade.

Mimi Bertillon

Works very hard in order fulfilment. Is involved in the essential part of the process where the orders are packed and addressed according to the instructions provided by Sanjay and Jake and their colleagues. Is allergic to oranges and marmalade.

Sam O'Farrell

General 'all rounder'. Has worked for the Millers for many years and is seen as the 'old timer'. Has experience of all areas of the marmalade business (and the farm as well). Is currently looking after the production line from cleaning the jars to filling them. Affectionately known as 'Grumpy'

Risk Exercise

Working in small groups, explore the following aspects of risk:

For this project as you currently understand it, use the 8 Principles and the Project Approach Questionnaire (PAQ) to identify risk. To ensure your team understands how the PAQ is used, each group member should select a few statements from different parts of the questionnaire.

- Identify your group's 3 biggest risks to using an Agile approach. For each of your top risks identify mitigation actions for them.

- For balance, also be prepared to talk about the key risks of choosing a traditional approach to project management on this project.

***Exercise objectives**:*

Successful Agile is about preparing as much as possible in the early stages. Often the situation is less than perfect, and being clear where the risks lie and how these may be managed is key.

Project Approach Questionnaire for the Case Study Exercise

Project:	Name :
Date :	Position :

DSDM Atern Project Approach Questionnaire		***Indicate the closest collective opinion***					***Where appropriate, comment on issues or risks related to a more negative response to this aspect of the DSDM Atern approach***
Ref	**Statement**	**Strongly Agree**	**Agree**	**Neutral**	**Disagree**	**Strongly Disagree**	
1	The business driver behind the project is clearly stated and is visible to all members of the project team						
2	The Business Sponsor and/or Business Visionary demonstrate clear ownership of the project.						
3	The Business Sponsor, Business Visionary and Technical Coordinator all understand and accept the DSDM Atern philosophy						
4	The development has a clearly-defined timescale						
5	The requirements can be prioritised and there is flexibility to accept that not all requirements are 'Must Have' requirements						
6	Requirements have been defined at a high level at the outset of the project, and it is acknowledged that changes are likely during development						
7	It is accepted that the detail of both the requirements and the solution will emerge as the project progresses						

DSDM Atern Project Approach Questionnaire		*Indicate the closest collective opinion*					*Where appropriate, comment on issues or risks related to a more negative response to this aspect of the DSDM Atern approach*
Ref	**Statement**	**Strongly Agree**	**Agree**	**Neutral**	**Disagree**	**Strongly Disagree**	
8	The Business Sponsor and Business Visionary are aware of the importance of active business involvement and have the willingness and authority to commit appropriate business resources to the project as required.						
9	The Business Ambassadors are sufficiently empowered to guide the day to day evolution of the solution						
10	The Solution Developers are sufficiently empowered to provide the best solution they can from a business perspective within pre-agreed architectural constraints.						
11	Solution development resources are allocated at an appropriate level and the team will be largely stable throughout the project (or, at least, throughout each increment).						
12	The Project Team and Solution Development team are able to adopt the roles and responsibilities within DSDM Atern						
13	It will be possible for the Solution Developers to have easy access to Business Ambassadors and Business Advisors throughout the project.						
14	The Solution Development team (including both business and solution development resources) will have the appropriate collective knowledge and/or technical skills to deliver the solution.						

DSDM Atern Project Approach Questionnaire		*Indicate the closest collective opinion*					*Where appropriate, comment on issues or risks related to a more negative response to this aspect of the DSDM Atern approach*
Ref	**Statement**	**Strongly Agree**	**Agree**	**Neutral**	**Disagree**	**Strongly Disagree**	
15	The Solution Development team (including both business and solution development resources) will have the appropriate soft skills (communication, negotiation etc.) to work effectively with each other and those around them.						
16	Strategies for continuous communication and collaborative working practices are sufficient to clearly support iterative development of the solution						
17	The development technology tools and techniques support an iterative approach to solution development						
18	There are no technical, contractual or other constraints to prevent the solution being broken into increments for development and delivery (even if the products of such increments are not deployed immediately)						
19	All project participants understand and accept that on-time delivery of an acceptable solution is the primary measure of success for the project						
20	All parties accept that continual assessment of the fitness for purposes of all deliverables during development is essential						
21	There are no mandatory standards or practices in force that will work against the evolution of a solution from a baselined set of high level requirements						

Communication Exercise - Communicating with Stakeholders

Working in small groups:

- Run a small facilitated workshop :
- Some of you will take the business roles identified earlier and possibly other team roles (your trainer will advise)
- One of your group will be the facilitator
- **Outcome:**
  after exploring the specific communication issues with these stakeholders produce a list of your group's top 5 ways to ensure successful communication with them.

Discuss agile communication for all stakeholders involved in the project. Consider the following during discussion: location, frequency of communication, communication medium, information needed, reporting, decision making, numbers of people involved, types of community, types of information etc

Further conversations have identified the following additional stakeholders the project will need to involve:

Health and Safety (H&S) Auditors – Millers Marmalade has to be audited annually to make sure that they comply with H&S standards for marmalade production. They also have to be able to provide local language ingredient and allergy information in all countries where their marmalade is marketed.

John Jones – Television Producer responsible for the '100 Best Food Websites' programme. His 3 researchers are planning their approach to finding and checking food websites. They will be short-listing the top 500 for more detailed investigation in months 4 and 5, with final production during month 6

Mary Hannington – a manager with TestyToo (reposible for this account). TestyToo are an agile testing company and have been appointed by your management to work with your company to perform testing. They are very familiar with Agile. Although Mary is reasonably local to you, many of the testers are based offshore – with an 8 hour time difference.

Jeanne Haute – marketing director for the Orange D'Or Award and very specific about how award winners promote the Award in their marketing materials. Has been known to take to court anyone breaking the Terms and Conditions of the brand. Also very particular about the look and feel of the websites where the award logo appears – doesn't want anything 'low quality'.

***Exercise objectives**:*

Improving communication is a key aspect of project success, and particularly important when using Agile. This exercise is designed to prompt thought about a variety of important communication considerations.

Prioritising Requirements Exercise

Working in small groups:

Decide on your group's 'MoSCoW' for each of the requirements provided.

Remember that Must Have requirements will create your *Minimum Usable Subset* for an early release. Without all the Must Haves, it would be unviable / unsafe to release anything. The difference between Should Haves and Could Haves will be in the value or impact, or the difficulty / ease of a workaround.

How close is your output to the 60:20:20 guideline for effort involved?
What value does each of the requirements add to the overall business objective?

For this exercise, some group members should take the role of the Business Ambassador and Business Visionary to ensure that the MoSCoW prioritisation is business driven. The others should take other roles.

Where you are unsure about a business related question, your trainer will provide an answer, alternatively you may make an assumption so that you can move on quickly.

Exercise objectives:

As the delivery and balance of the MoSCoWed Prioritised List is essential to the success of an Agile project, it is important to understand prioritisation. Also it is useful to have experienced prioritisation from perspectives.other than your usual job role.

Prioritisation Exercise - Millers Marmalade

REQUIREMENTS LIST FOR THE NEW WEB SITE (Total 62 days effort)

No.	Requirement	MSCW	Estimate
	As a <role> I want <something> so that <outcome>		
1	As a **customer** I want **to order marmalade** so that **I can specify what I want to receive**		2
2	As a customer I want to change an existing open order (i.e. size, amount) so that I can ensure my order is what I actually want		3
3	As a customer I want to remove some marmalade from an existing order so that I can ensure my order is what I actually want		4
4	As a customer I want to choose to have the order gift wrapped so that I can send it as a present already-wrapped to someone else		2
5	As a customer I want to browse product information (e.g. ingredients) so I can make sure the ingredients are healthy / will not cause an allergic reaction		5
6	As a customer I want to see a price list/catalogue of marmalades so I know what my order would cost		2
7	As an I.T. manager I want to ensure that the transaction over the web is secure so that customers have confidence in using our website		3
8	As a customer I want to search and filter by size, price and type of marmalade so I can quickly search for the exact item(s) I want to order		4
9	As a Customer Services Manager I want to allow a visitor to sign up to receive a newsletter so that we can provide more information in order to encourage further sales		2
10	As a Customer Services Manager I want to have a customer feedback facility so that we can act on negative comments quickly		1
11	As a Head of Operations I want to promote special offers so that we can easily draw these to our customers attention		2
12	As a customer I want to be able to enter a separate invoice address so that I can send marmalade to addresses other than my own		4
13	As a customer I want to be able to enter alternative delivery addresses so that I can get my orders delivered to friends and family, and to my workplace		1

14	As an Owner of Millers I want to provide information about Millers ('about us') so that we can promote the Millers family image		1
15	As an Owner of Millers I want to allow payment by credit card/debit card so that we avoid cheque handling, which will cost us more money per transaction		4
16	As a Head of Operations I want to automatically authorise credit card payment so that the order payment process needs the least possible number of staff		3
17	As a customer I want to have other payment methods (e.g. cheque) so that I can buy marmalade even if I do not have a credit / debit card		2
18	As a Fulfilment Manager I want to notify the customer that the order has been dispatched so that the customer knows their order is on its way		2
19	As a customer I want to create, view and amend my customer details so that my details are always correct		5
20	As a customer I want to delete my customer details so that I will no longer receive marketing information from Millers		2
21	As a Head of Account Management I want to provide a trade area sign on facility so that my large order customers can process their orders as quickly as possible		2
22	As a customer I want to receive discounts related to size of order so that I can see I am getting good value on larger orders		3
23	As a customer I want to have a choice of delivery times so that I can be sure I am at home to take delivery		3

Delivery Planning Exercise

Working in small groups :

Using your MoSCoWed requirements from the previous exercise, create a Delivery Plan (using sticky notes on the table or wall) for the website element of the project. Define a number of Increments and specify the objective of each increment.

***Exercise objectives**:*

To demonstrate the style of Agile planning. Agile Planning should be product based, aimed at showing what products will be produced when, and who will be involved in the product. Agile plans are normally created collaboratively, rather than as solitary exercise Project Manager exercise. Sticky note planning is a common style of achieving a plan, as it allows engagement and buy-in from the whole group.

Additional Practicals – for reflections on implementing Agile

Creating an Agile Briefing

The following can be used as an additional exercise for Millers Marmalade or for individual reflection by simply reviewing the areas for your own environment

The senior managers at Food World have requested a briefing / short presentation on Agile to see if it is of interest to them. This briefing should last for no more than 5 minutes and it should outline the benefits of using Agile Project Management.

Choose a presentation to cover one of the following concerns around Agile:

- "How can we prioritise? All our requirements are Must Haves"
- "I want to know exactly what I am getting and how much it will cost"
- "Our business people are far too busy to work this way!"
- "I've heard that agile is a very risky way of working."
- "People tell me that going agile means lowering quality"
- "How will going agile address our planning and resourcing issues?"
- "If we go agile, what's in it for me?"
  (position this from various roles i.e. different points of view)

---------- some additional reflections for your environment ---------

Consider how the following will be affected by agile:

Testing, Configuration Management, Change Management,

Reporting, Measurements (Key Performance Indicators / progress etc)

Working environment, resources (not just people)

Communication mechanisms

How will you create effective empowerment, who needs to be involved?

What would you look for in the business people involved? And how would you ensure the correct levels of involvement?